Homework Busters – Final Team Presentation

School solutions for today and tomorrow

AnonReview

A reddit and stackoverflow inspired platform meant for students to receive peer critique from other students

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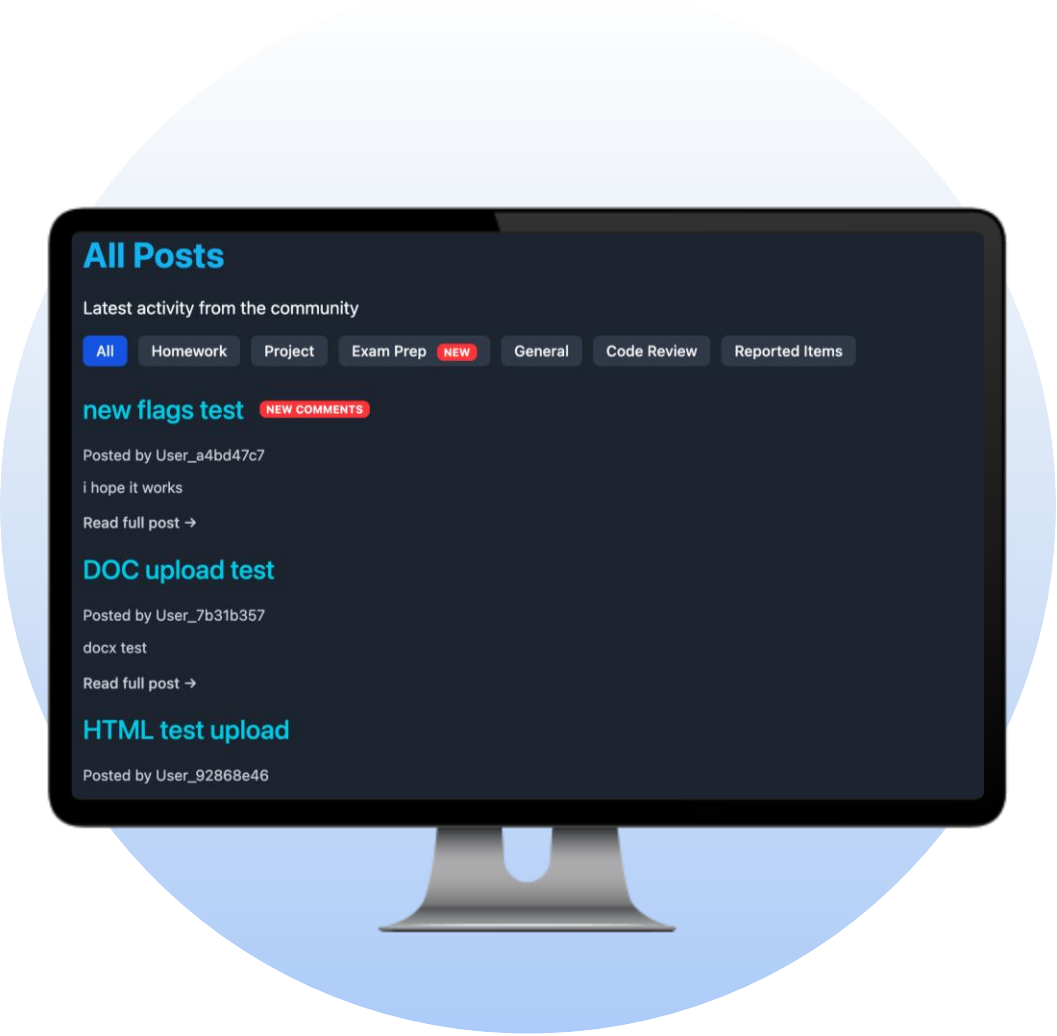
Course: Software Development Tools and Methods

Problem Motivation

Improve student to student interaction during online schooling



Anon Review Product Overview

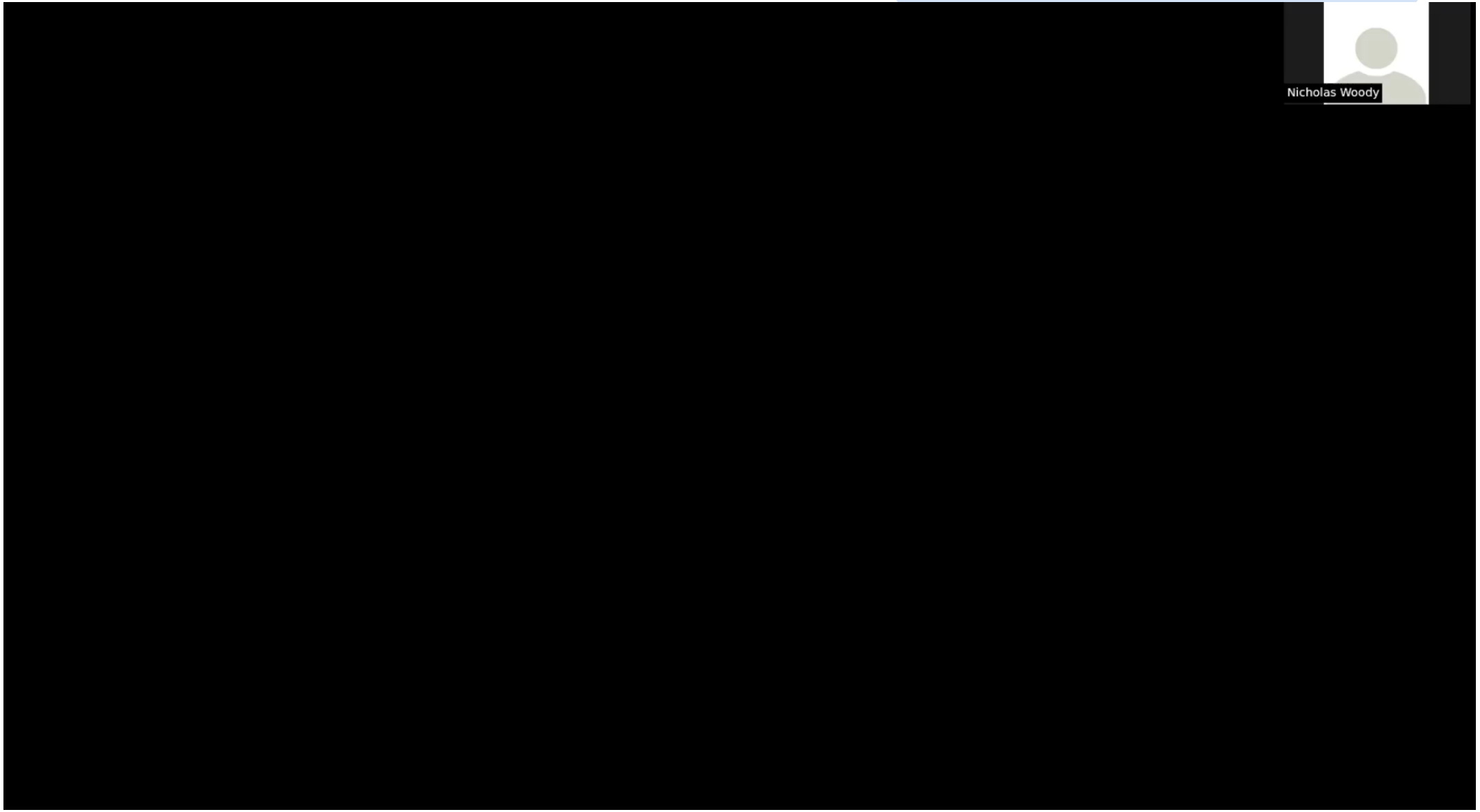
AnonReview is a web-based peer review platform that allows students to anonymously submit homework and exam related content for feedback. Submissions can include written responses, code snippets, and PDF documents. The platform supports inline comments and small group review so students can receive targeted help and collaborate with peers in a low pressure environment.

Demo Flow Overview

•Our live demo will follow this sequence:

1. User registration and login
2. Creating a post
3. Commenting and voting
4. Flagging posts
5. Dashboard Interaction
6. Admin interactions

Demonstration



Core Technologies

- Frontend: HTML, CSS, Jinja2
  - Purpose: Webpage display, styling, and control flow
- Backend: Flask
  - Purpose: Lightweight Web Server Gateway Interface (WSGI) app framework
- Database: SQLite4 (development)
  - Purpose: Implements a lightweight SQL database engine
- Database: PostgresSQL (production)
  - Purpose: Advanced Relational Database Management System (RDBMS)

Libraries and Dependencies

- Flask-WTForms
  - Purpose: Integrates Flask and the WTForms library. For easy form creation, validation, and rendering
- Flask-Login
  - Purpose: Handles user and session management for Flask applications
- Flask-Security (passlib)
  - Purpose: Provides security mechanisms for Flask applications. Ours mainly uses the passlib library it includes (for password hashing and verification)

Libraries and Dependencies (continued)

- BeautifulSoup4
  - Purpose: Web scraping library, pulls data from HTML and XML files. Used for posts in our application.
- PyPDF2
  - Purpose: Library for easy PDF handling (editing, securing, text retrieval). Used for posts in our application.
- python-docx
  - Purpose: Library for creating and updating MS Word files. Used for posts in our application.
- python-dotenv
  - Purpose: Library for reading and setting environment variables

Workflow

- GitHub
  - Purpose: Home of the AnonReview repository, along with our weekly update and project milestone documents
- Trello (on Jira)
  - Purpose: Project tracking (essentially a checklist of our weekly tasks)

Unit Testing

- unittest
  - Purpose: Python library for creating and running unit tests. We used it to test our database methods.

Usefulness verdict for all tools:

- Nearly every tool we used was excellent in terms of usefulness.
- Each tool was either necessary for the application to run the way it does or made the development process much easier.
- The only tool that we felt was not entirely necessary was our project tracking tool, Trello- we found that our weekly meetings and status documents worked well enough to keep us on track.

Challenges & Mitigations

- **Communication** – Early on we found that pieces of our application often depended on other pieces. Often we'd have questions about other team member's code, or may need to change some code to make our piece of the application work. To help ease communication between team-members, we created a Teams chat for one-off questions and started a mid-week check in.

- **Database Planning** – The database needed to be changed many times throughout our project. Sometimes we'd need to add or change a field. Using DB methods whenever possible helped reduce the time it took to deploy these changes, and communicating any changes which were made in our meetings and in our chat helped us succeed.

- **Merging Conflicts** – Sometimes team members would be working on files at the same time or we'd be working on separate branches and needed to merge. Gaining knowledge of git merges was helpful to mitigate any issues we encountered when attempting to merge our changes.

Team Dynamics

- Each team member assigned themselves their own epic each week
- Each team member built on and tested their epic features
- Scrum masters rotated
- Each team member handled raising issues, blockers, and asking for help
- A progress update was sent via Teams every Thursday
- A stand up Every Monday night.

Future Work

If we continued we would have:

- Implemented breakout rooms for study groups and team collaboration
- Added functionality for image upload
- Allowed for user notifications
- Save documents outside on a cloud platform
- Deploy using our domain
- Remove the dashboard option from the index pages

Final Reflection

We learned how to:

- How to build a web application.
- Work more efficiently as a team using industry tools
- Collaborate via github and the tools it offers
- Build a CI/CD pipeline
- Expanded library knowledge arsenal

Thank you

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